Assessment 2 Q&A

SOST10142/20142 Applied Statistics for Social Scientists

Some Questions and Answers concerning Assessment 2 (data analysis report)

Q: What dependent variable should I use for my analysis? Can you just choose one for me?

We understand that it may be daunting to select a dependent variable from a real dataset and among many potential options. However, this decision is part of the assessment (i.e., selecting an appropriate variable to regress on a set of predictors); thus, we cannot provide you with a specific selection. We do, however, encourage you to select a variable that is roughly continuous (i.e., 6 or more ordered points), which means that you can likely model this type of outcome using Ordinary Least Squares (OLS) estimation. You are, of course, free to choose a dependent variable measured on a number of different response scales (e.g., binary, ordered categorical, count, nominal, etc.), and the expectation would be that you would apply the correct estimation approach in this case (i.e., a Generalized Linear Model with the appropriate link function). This decision may have implications for your interpretation of results. Whatever you decide, the most important thing to do is demonstrate that you understand how to model and interpret the survey data, while addressing any potential issues that you may encounter.

Q: What does a 'focal continuous predictor' mean?

The terminology can be explained using an example.

```
model2 <- lm(bmival ~ socialclass + age + gender +
             employed + fruitveg + alcohol, data = health)
```

In the above example, Body Mass Index (BMI) *is regressed on* several predictor variables -- social class, but also controlling for other factors including age, gender, employment status, fruit and vegetable intake, and alcohol consumption. We are particularly interested in the **social class** variable's regression coefficient (this is just to give you a focus in the research; otherwise, you will look at all six varaibles, leading to overly complicated discussions).

Q: How many variables do I need to pick? What type?

You will choose a dependent variable first. Then, you will choose five explanatory variables in total, with one being the focal predictor.

Q: Do I need to explain all the other choices of control variables?

Yes, you do need to justify why you choose those particular variables. You may link these to your research question, if helpful. Your arguments will be strengthened if you cite some literature, but remember that the focus of the report is on data analysis and interpret, not lengthy literature reviews.

Q: Are we expected to remove any outliers in the data or treat it as accurate?

Anything not mentioned in the Assessment Instructions document, you may treat it as optional. If you follow the instructions and include components that you are asked to, you'll pass. For anything else, you can use your judgment depending on your goals (the better able to demonstrate your grasp of concepts and ability to conduct data analysis, the higher the probability of earning extra marks, provided things are done correctly).

As such, treating outliers is not essential. You will not be penalised for not including this. If you do, it'll improve the quality of the report, potentially contributing to higher marks. Please note that a simple inclusion/exclusion of contents will not lead to certain grades. We will have to examine the overall *quality and coherence* of the full submitted report.

Q: What's the difference between References and a Bibliography.

References mean that you include citations in the text, for example, Smith (2008), as well as a list of full details of each in-text citation at the end of the report.

A bibliography usually means that you have consulted the source, but it does not appear in an in-text citation. As it only appears in the Bibliography, readers have no information at which point you have consulted this particular academic source.